```
def binary_search(arr, x):
    left = 0
    right = len(arr) - 1

    while left <= right:
        mid = (left + right) // 2

        if arr[mid] == x:
            return mid
        elif arr[mid] < x:
            left = mid + 1
        else:
            right = mid - 1

    return -1
n = int(input())
arr = list(map(int, input().split()))
x = int(input())
result = binary_search(arr, x)
print(result)
```